

Research Interests

My research focuses on **resource management and coordination in multi-tenant networks**, with an emphasis on balancing efficiency, fairness, and performance isolation. I design networked systems and coordination mechanisms that enable independently operated tenants to share constrained resources—such as spectrum and network capacity—despite conflicting performance objectives.

Education

University of Illinois, Urbana-Champaign

August 2022 – Present

PhD. in Computer Science

Urbana, IL

- Advised by: Radhika Mittal

Lahore University of Management Sciences (LUMS)

September 2018 – May 2022

Bachelor of Science in Computer Science

Lahore, Pakistan

- Placed on the Dean's Honors List for academic excellence for years 2020-2022

Industry Experience

Google

May 2026 – August 2026

PhD Software Engineering Intern – Hosts: Raymond Lam, Clayton Pence

Seattle, WA

- Designed a hybrid root-cause analysis and triage pipeline for cloud infrastructure, combining deterministic heuristics with a specialized agent to triage SLO violations.
- Optimized telemetry ingestion by using statistical anomaly filtering drastically reducing token usage & context bloat.
- Deployed as a completely automated pipeline with analytics dashboards and issue routing, eliminating manual triage.
- Grew automated diagnosis and triage coverage from 0% to >80% of daily bugs, significantly reducing manual toil for team engineers.

Google

May 2025 – August 2025

PhD Software Engineering Intern – Hosts: Yingjie Bi, Hamid Bazzaz

Sunnyvale, CA

- Implemented a simulation framework to estimate reservable bandwidth for Premium Bandwidth in datacenter clusters.
- Designed a max-min fairness model to ensure fair and efficient bandwidth allocation across competing services.
- Integrated the simulator with large-scale cluster management frameworks, leveraging Borg production checkpoints and historical usage data from Rumbo for accurate modeling.

Publications

Saving Private Cellular: Addressing The Coordination Vacuum in CBRS

Taimoor Tariq, Radhika Mittal

HotMobile 2026

Performance Isolation for 5G RAN Slicing Across Multiple Interfering Cells

Taimoor Tariq, Yongzhou Chen, Haitham Hassanieh, Radhika Mittal

NINeS 2026

5G RAN Slicing with Load Balanced Handovers

Yongzhou Chen, Taimoor Tariq, Haitham Hassanieh, Radhika Mittal

arXiv 2025

Centralized Traffic Engineering for Networked Farm Applications

Ammar Tahir, Yueshen Li, Jianli Jin, Changxin Zhang, Daniel Moon, Aganze Mihigo, Taimoor Tariq, Deepak Vasisht, Radhika Mittal

IEEE / ACM SEC 2025

Enabling Emerging Edge Applications Through a 5G Control Plane Intervention

Mukhtiar Ahmad, Muhammad Nawazish Ali, Taimoor Tariq, Basit Iqbal, Taqi Raza, Zafar Ayyub Qazi

ACM CoNEXT 2022

Neutrino: A Fast and Consistent Edge-based Cellular Control Plane

Mukhtiar Ahmad, Muhammad Nawazish Ali, **Taimoor Tariq**, Syed Usman Jafri, Adnan Abbass, Syeda Mashal Abbas, Basit Iqbal, Zartash Uzmi, Zafar Ayyub Qazi

IEEE/ACM Transactions on Networking, 2022

Fast Failure Detectors for 5G Edge Deployments (Poster)

Taimoor Tariq, Maha Kamal, Vafa Batool

Student Research Competition, SOSP 2021

Teaching Experience

Teaching Assistant

CS-425: Distributed Systems

August 2024 – December 2024

UIUC, IL

Teaching Assistant

CS-678: Topics in Internet Research (Graduate)

January 2022 – May 2022

LUMS, Pakistan

Lead Teaching Assistant

CS-582: Distributed Systems (Graduate)

September 2021 – December 2021

LUMS, Pakistan

Teaching Assistant

CS-382: Network-Centric Computing (Undergraduate)

January 2021 – May 2021

LUMS, Pakistan